

Experiment No:3

DDL Queries

Q1. Create following tables for bank database with specific constraint

Branch (branchname, branchcity, asset)

- city and asset should not null, asset greater than 0

Customer (custname, custstreet, custcity)

- city not null

Account (accnum, branchname, balance)

- balance greater than 500

Loan (loanno, branchname, amount)

- amount greater than 0

Create a relation borrower which is between customer and loan

Create a relation Depositor which is between customer and accnum

Perform following:

1. Add Constraint street not null
2. Add a column phoneNo to customer table.
3. Add column custage to customer table
4. Add Constraint age greater than 18
5. Change the size of the phoneNo to varchar(10).
6. Drop the column phoneNo from customer table.
7. Remove the constraint on age.

create table Branch(branchname varchar(40) primary key,branchcity varchar(50) not null,asset float not null check(asset>=0));

desc Branch;

	Field	Type	Null	Key	Default	Extra
►	branchname	varchar(40)	NO	PRI	NULL	
	branchcity	varchar(50)	NO		NULL	
	asset	float	NO		NULL	

create table Customer(custname varchar(50) primary key,custstreet varchar(40),custcity varchar(40) not null);

describe Customer;

	Field	Type	Null	Key	Default	Extra
►	custname	varchar(50)	NO	PRI	NULL	
	custstreet	varchar(40)	NO		NULL	
	custcity	varchar(40)	NO		NULL	

create table Account(accnum varchar(25) primary key,branchname varchar(50),balance float check(balance>500),

constraint branchname_fk foreign key(branchname)

references Branch(branchname) on delete cascade on update cascade);

describe Account;

	Field	Type	Null	Key	Default	Extra
►	accnum	varchar(25)	NO	PRI	NULL	
	branchname	varchar(50)	YES	MUL	NULL	
	balance	float	YES		NULL	

create table Loan(loanno varchar(30) primary key,branchname varchar(40),amount decimal(10,2) check(amount>0),

constraint bname_fk foreign key(branchname)

references Branch(branchname) on delete cascade on update cascade);

describe Loan;

	Field	Type	Null	Key	Default	Extra
►	loanno	varchar(30)	NO	PRI	NULL	
	branchname	varchar(40)	YES	MUL	NULL	
	amount	decimal(10,2)	YES		NULL	

```
create table Borrower(loanno varchar(30),custname varchar(40),
```

```
constraint loanno_fk foreign key (loanno) references Loan(loanno) on delete cascade on  
update cascade,
```

```
constraint custname_fk foreign key(custname) references Customer(custname) on delete  
cascade on update cascade);
```

```
describe Borrower;
```

	Field	Type	Null	Key	Default	Extra
▶	loanno	varchar(30)	YES	MUL	NULL	
	custname	varchar(40)	YES	MUL	NULL	

```
create table Depositor(accnum varchar(30),custname varchar(40),
```

```
constraint accnum_fk foreign key (accnum) references Account(accnum) on delete cascade  
on update cascade,
```

```
constraint cname_fk foreign key(custname) references Customer(custname) on delete  
cascade on update cascade);
```

```
describe Depositor;
```

	Field	Type	Null	Key	Default	Extra
▶	accnum	varchar(30)	YES	MUL	NULL	
	custname	varchar(40)	YES	MUL	NULL	

```
1.alter table Customer modify custstreet varchar(40) not null;
```

```
desc Customer;
```

	Field	Type	Null	Key	Default	Extra
▶	custname	varchar(25)	NO	PRI	NULL	
	custstreet	varchar(25)	YES		NULL	
	custcity	varchar(25)	NO		NULL	

```
2.alter table Customer add phoneNo int;
```

```
desc Customer;
```

	Field	Type	Null	Key	Default	Extra
▶	custname	varchar(50)	NO	PRI	NULL	
	custstreet	varchar(40)	NO		NULL	
	custcity	varchar(40)	NO		NULL	
	custage	int	YES		NULL	
	phoneNo	int	YES		NULL	

3.alter table Customer add constraint age check(custage>18);

desc Customer;

	Field	Type	Null	Key	Default	Extra
▶	custname	varchar(50)	NO	PRI	NULL	
	custstreet	varchar(40)	NO		NULL	
	custcity	varchar(40)	NO		NULL	
	custage	int	YES		NULL	
	phoneNo	int	YES		NULL	

4.alter table Customer modify phoneNo varchar(10);

desc Customer;

	Field	Type	Null	Key	Default	Extra
▶	custname	varchar(50)	NO	PRI	NULL	
	custstreet	varchar(40)	NO		NULL	
	custcity	varchar(40)	NO		NULL	
	custage	int	YES		NULL	
	phoneNo	varchar(10)	YES		NULL	

5.alter table Customer drop column phoneNo;

desc Customer;

	Field	Type	Null	Key	Default	Extra
▶	custname	varchar(50)	NO	PRI	NULL	
	custstreet	varchar(40)	NO		NULL	
	custcity	varchar(40)	NO		NULL	
	custage	int	YES		NULL	

6.alter table Customer drop constraint age;

describe Customer;

	Field	Type	Null	Key	Default	Extra
▶	custname	varchar(50)	NO	PRI	NULL	
	custstreet	varchar(40)	NO		NULL	
	custcity	varchar(40)	NO		NULL	
	custage	int	YES		NULL	

DML Queries –

1. Insert at least 5 rows in each table

```
insert into Branch(branchname,branchcity,asset) values('SBI','Gadhinglaj',150000.00),
('BOI','Ichalkaranji',200000.00),
('BOB','Sangali',250000.00),
('Ajara Urban','Kolhapur',180000.00),
('KDC','Ajara',10000.00);
```

```
select * from Branch;
```

	branchname	branchcity	asset
▶	Ajara Urban	Kolhapur	180000
	BOB	Sangali	250000
	BOI	Ichalkaranji	200000
	KDC	Ajara	10000
	SBI	Gadhinglaj	150000

```
insert into Customer(custname,custstreet,custcity) values
```

```
('Gayatri','MG road','Gadhinglaj'),
('Pranali','FC road','Ichalkaranji'),
('Pramila','Vardarge road','Sangali'),
('Sanika','Market yard','Kolhapur'),
('Sneha','Parel','Chandgad');
```

```
Select * from Customer;
```

	custname	custstreet	custcity	custage
▶	Gayatri	MG road	Gadhinglaj	NULL
	Pramila	Vardarge road	Sangali	NULL
	Pranali	FC road	Ichalkaranji	NULL
	Sanika	Market yard	Kolhapur	NULL
	Sneha	Parel	Chandgad	NULL

```
insert into Account(accnum,branchname,balance) values('AC101','SBI',15000.00),
('AC102','BOI',20000.00),
('AC103','BOB',25000.00),
('AC104','Ajara Urban',18000.00),
('AC105','KDC',10000.00);
```

select * from Account;

	accnum	branchname	balance
▶	AC101	SBI	15000
	AC102	BOI	20000
	AC103	BOB	25000
	AC104	Ajara Urban	18000
	AC105	KDC	10000

```
insert into Loan(loanno,branchname,amount) values('LN101','SBI',20000.00),
('LN102','BOI',22000.00),
('LN103','BOB',28000.00),
('LN104','Ajara Urban',34000.00),
('LN105','KDC',50000.00);
```

select * from Loan;

	loanno	branchname	amount
▶	LN101	SBI	20000.00
	LN102	BOI	22000.00
	LN103	BOB	28000.00
	LN104	Ajara Urban	34000.00
	LN105	KDC	50000.00

```
insert into Borrower(loanno,custname) values('LN101','Gayatri'),
('LN102','Pranali'),
('LN103','Pramila'),
('LN104','Sanika'),
('LN105','Sneha');
```

select * from Borrower;

	loanno	custname
▶	LN101	Gayatri
	LN102	Pranali
	LN103	Pramila
	LN104	Sanika
	LN105	Sneha

```

insert into Depositor(accnum,custname) values('AC101','Gayatri'),
('AC102','Pranali'),
('AC103','Pramila'),
('AC104','Sanika'),
('AC105','Sneha');

```

	accnum	custname
▶	AC101	Gayatri
	AC102	Pranali
	AC103	Pramila
	AC104	Sanika
	AC105	Sneha

2. Insert an customer vivek who live in street is main street and city is Ichalkarnji

```

insert into Customer values('Vivek','Main Street','Ichalkaranji',null);

```

```

select * from Customer;

```

	custname	custstreet	custcity	custage
▶	Gayatri	MG road	Gadhinglaj	NULL
	Pramila	Vardarge road	Sangali	NULL
	Pranali	FC road	Ichalkaranji	NULL
	Sanika	Market yard	Kolhapur	NULL
	Sneha	Parel	Chandgad	NULL
	Vivek	Main Street	Ichalkaranji	NULL

4. Change the assests of Perryridge branch to 3400

```

update Branch set asset=3400 where branchname='KDC';

```

```

select * from Branch;

```

	loanno	branchname	amount
▶	LN101	SBI	20000.00
	LN102	BOI	22000.00
	LN103	BOB	28000.00
	LN104	Ajara Urban	34000.00
	LN105	KDC	50000.00

5. Add 2% interest to all bank account balances with a balance of 500 or less

```

update Account set balance=balance+(balance*0.02) where balance<=25000 ;

```

```

select * from Account;

```

	accnum	branchname	balance
▶	AC101	SBI	15300
	AC102	BOI	20400
	AC103	BOB	25500
	AC104	Ajara Urban	18360
•	NULL	NULL	NULL

6. Transfer the accounts and loans of Perryridge branch to Downtown branch

7. Transfer Rs. 100 from account AC101 to AC102.

update Account set balance=case when accnum='AC101' then balance-100 when accnum='AC102' then balance+100 else balance end;

select * from Account;

	accnum	branchname	balance
▶	AC101	SBI	15200
	AC102	BOI	20500
	AC103	BOB	25500
	AC104	Ajara Urban	18360

8. Delete the branch 'KDC'

delete from Branch where branchname='KDC';

select * from Branch;

	branchname	branchcity	asset
▶	Ajara Urban	Kolhapur	180000
	BOB	Sangali	250000
	BOI	Ichalkaranji	200000
	SBI	Gadhinglaj	150000

9. Remove all the customer who live in "Sangali"

delete from Customer where custcity='Sangali';

select * from Customer;

	custname	custstreet	custcity	custage
▶	Gayatri	MG road	Gadhinglaj	NULL
	Pranali	FC road	Ichalkaranji	NULL
	Sanika	Market yard	Kolhapur	NULL
	Sneha	Parel	Chandgad	NULL
	Vivek	Main Street	Ichalkaranji	NULL

10. Remove all the accounts

delete from Account;

select * from Account;

	accnum	branchname	balance
•	NULL	NULL	NULL

11. List all branch names and their assests

select branchname,asset from Branch;

	branchname	asset
▶	Ajara Urban	180000
	BOB	250000
	BOI	200000
	SBI	150000

12. List all loans with amount greater than 1000.

select * from loan where amount>1000;

	loanno	branchname	amount
▶	LN101	SBI	20000.00
	LN102	BOI	22000.00
	LN103	BOB	28000.00
	LN104	Ajara Urban	34000.00

13. Find bank accounts with a balance under 2000

select * from Account where balance<20000;

	accnum	branchname	balance
▶	AC101	SBI	15000
	AC104	Ajara Urban	18000

14. List all accounts of BOI branch with balance less than 30000

select * from Account where branchname='BOI' and balance<30000;

	accnum	branchname	balance
▶	AC102	BOI	20000

15. Find the names of all branches in the loan relation

select distinct branchname from Loan;

	branchname
▶	Ajara Urban
	BOB
	BOI
	SBI

16. Find all loan numbers for loans made at the SBI branch with loan amounts greater than Rs. 10000.

```
select loanno from Loan where branchname='SBI' and amount>10000;
```

	loanno
▶	LN101

17. Find the loan number of those loans with loan amounts between Rs. 10000 and Rs. 50000

```
select loanno from Loan where amount between 10000 and 50000;
```

	loanno
▶	LN101
	LN102
	LN103
	LN104

18. Find the names of all customers.

```
select custname from Customer;
```

	custname
▶	Gayatri
	Pramila
	Pranali
	Sanika
	Sneha
	Vivek

19. Find the names of all branches in the loan relation, don't display duplicates.

```
select distinct branchname from loan;
```

	branchname
▶	Ajara Urban
	BOB
	BOI
	SBI

20. Display the entire Branch table.

select * from Branch;

	branchname	branchcity	asset
▶	Ajara Urban	Kolhapur	180000
	BOB	Sangali	250000
	BOI	Ichalkaranji	200000
	SBI	Gadhinglaj	150000

21. Find the account number for all accounts where the balance is greater than \$700.

select accnum from Account where balance>700;

	accnum
▶	AC101
	AC102
	AC103
	AC104

22. Find the account number and balance for all accounts from Brighton where the balance is greater than 10000.

select accnum,balance from Account where branchname='Ajara Urban' and balance>10000;

	accnum	balance
▶	AC104	18000

23. Display the branch name and assets from all branches in thousands of dollars and rename the assets column to 'assets in thousands'.

24. Find the name of all branches with assets between one and four million dollars.

25. Find the names of all customers whose street address includes the substring 'road'.

select custname from Customer where custstreet like '%road%';

	custname
▶	Gayatri
	Pramila
	Pranali

26. List all Accounts where the Bank_Branch begins with a 'C' and has 'a' as the second character

select * from account where branchname like 'Aj%';

	accnum	branchname	balance
▶	AC104	Ajara Urban	18000

27. List all Accounts where the Bank_Branch column has 'j' as the second character and has at least 3 character.

select * from Account where branchname like '_j_%';

	accnum	branchname	balance
▶	AC104	Ajara Urban	18000

28. Find all loan numbers that appear in the loan relation with null values for amount.

select loanno, amount from Loan where amount is null;

	loanno	amount
*	NULL	NULL